# Smart Swarm Robotic System for Earthquake Search and Rescue




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# Project Framework and Overview



- Earthquake-damaged environments can be unknown and difficult to search systematically.
- Two simulated robots cooperate to explore, map, share information, and plan a route to a virtual target.



**Unknown Environment** → **LiDAR Sensing** → **Occupancy Grid Mapping** → **Cooperative Exploration** → **Map and Data Sharing** → **Virtual Target Localization** → **Rescue-Route Planning**

**This work studies coordinated exploration and route planning in an initially unknown two-dimensional damaged environment, with practical relevance to structured search support.**

# Literature Review: Three Selected Papers



## Jadeja et al. (2024)

‘Survivor Detection Approach for Post Earthquake Search and Rescue Missions Based on Deep Learning Inspired Algorithms’

Deep-learning survivor detection; Faster R-CNN, SSD, YOLOv8, and YOLOv10; provides detection context, while this project uses a predefined virtual target.



## Pongsirijinda et al. (2024)

‘Distributed Multi-Robot Potential-Field-Based Exploration with Submap-Based Mapping and Noise-Augmented Strategy’

Distributed exploration using potential fields and submap-based collaborative mapping; informs cooperative exploration and map sharing.



## Calzolari et al. (2024)

‘Reinforcement Learning Driven Multi-Robot Exploration via Explicit Communication and Density-Based Frontier Search’

Multi-robot exploration using explicit communication, reinforcement learning, and density-based frontier search; informs coordination and information exchange.

**The proposed project focuses on simulated LiDAR, occupancy grids, frontier exploration, and communication.**

# Comparative Table of Previous Studies



| Research Paper | Main Techniques | Strengths | Contribution to Our Project |
|---|---|---|---|
| **Jadeja et al. (2024)** | Faster R-CNN, SSD, YOLOv8, YOLOv10 | Addresses deep-learning survivor detection | Helps define the distinction between visual survivor detection and our simulation-based target localization. |
| **Pongsirijinda et al. (2024)** | Distributed exploration, potential fields, submap mapping, noise augmentation | Connects multi-robot exploration with collaborative mapping | Inspires our approach to cooperative exploration and shared-map construction between two robots. |
| **Calzolari et al. (2024)** | Explicit communication, reinforcement learning, density-based frontier search | Connects exploration and communication | Informs our use of frontier-based exploration and robot communication, with greedy task allocation for target assignment. |

# Main and Secondary Project Objectives

## Main Objective

To design and simulate a cooperative two-robot system for exploration, mapping, and rescue-route planning in a simulated earthquake-damaged environment.

## Secondary Objectives

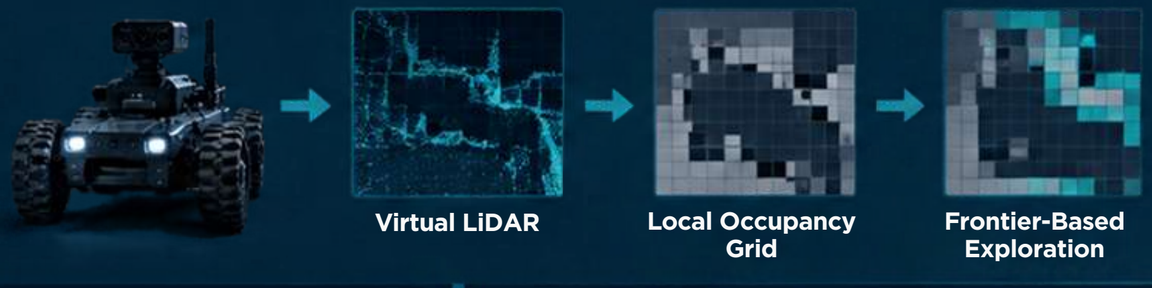
1. Develop a two-dimensional simulated disaster environment.
2. Implement simulated LiDAR sensing and occupancy grid mapping.
3. Apply frontier-based exploration and greedy task allocation.
4. Support communication and information sharing between the robots.
5. Implement A* path planning and replanning.
6. Represent a missing person using a predefined virtual target and a distance threshold.
7. Evaluate exploration coverage, redundant searching, mapping, and rescue-route planning.



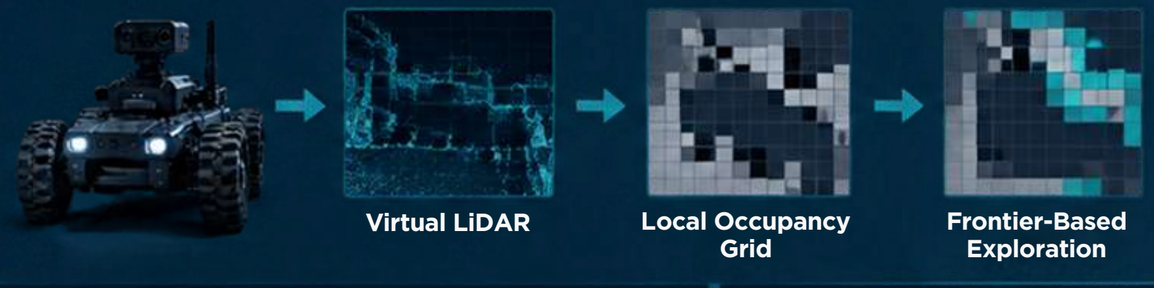
# Proposed System Architecture




**Robot 1: Virtual LiDAR → Local Occupancy Grid → Frontier-Based Exploration**




**Robot 2: Virtual LiDAR → Local Occupancy Grid → Frontier-Based Exploration**




**Communication and Information Exchange**

**Greedy Task Allocation**



**Shared Map Integration**



**Predefined Virtual Target Localization**



**A* Rescue-Route Planning**

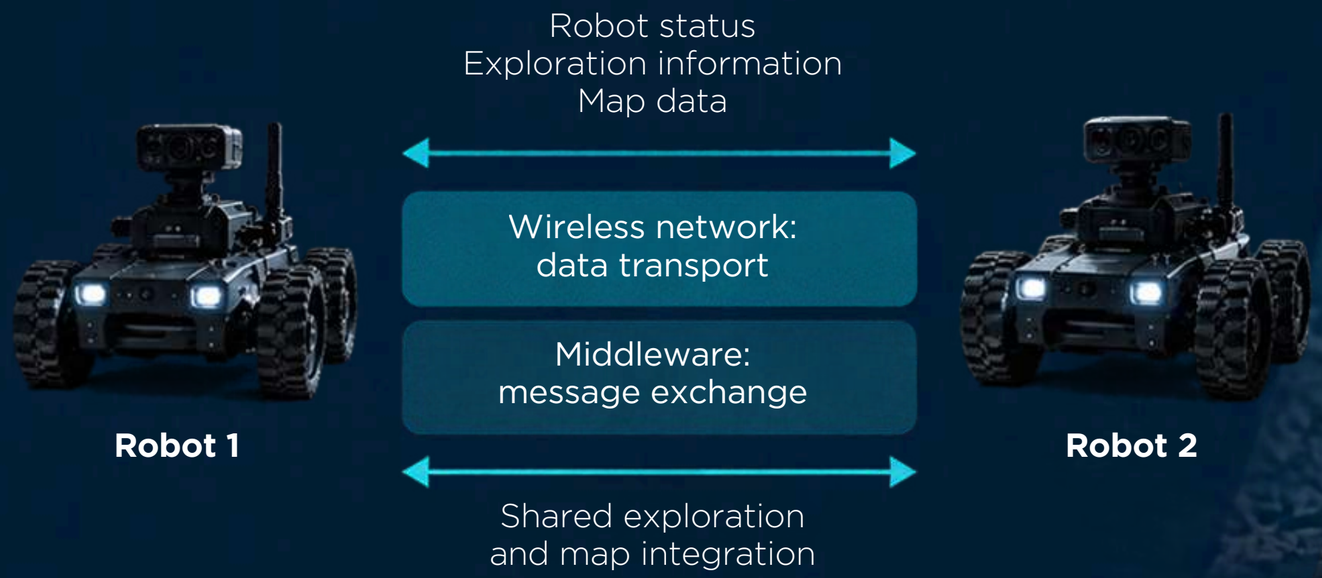


Proposed architecture; implementation status is not asserted.

# Technical Specifications

| Component | Proposed Specification | Purpose |
|---|---|---|
| **Robotic Platform** | Two cooperative ground robots | Cooperative exploration |
| **Operating Environment** | Two-dimensional earthquake-damaged simulation | Search scenario |
| **Sensing System** | Simulated LiDAR | Acquire virtual range data |
| **Mapping Method** | Occupancy grid mapping | Represent explored and unknown space |
| **Exploration Strategy** | Frontier-based exploration | Guide exploration toward unknown boundaries |
| **Task Allocation** | Greedy task allocation | Assign exploration tasks |
| **Path Planning** | A* algorithm with replanning where required | Plan and update routes |
| **Communication Architecture** | ROS 2 Topics and DDS/RTPS, subject to confirmation of the final implementation | Exchange robot information |
| **Information Exchange** | Robot status, exploration information, and map data as required by the final design | Support coordination |
| **Map Integration** | Combining shared mapping information | Create a shared map representation |
| **Target Representation** | Predefined virtual target representing a missing person, detected using a distance threshold | Represent target localization in simulation |
| **Simulation Scope** | Software simulation; no physical robot deployment | Define project boundaries |
| **Performance Evaluation** | Exploration coverage, redundant exploration, mapping performance, coordination, and rescue-route planning | Assess system behavior |

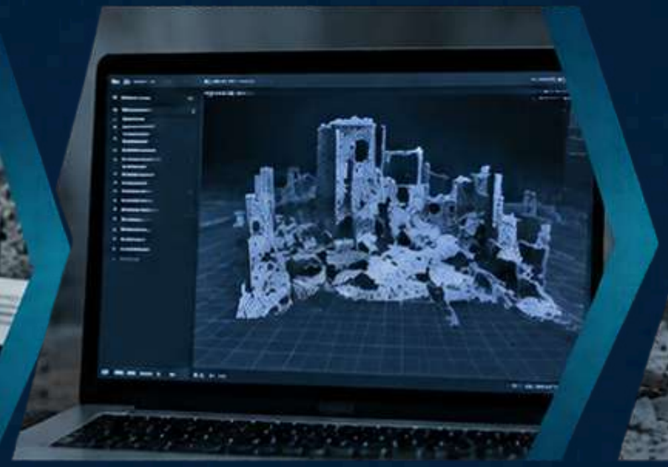
# Communication System and Coordination




- Information exchange supports coordinated exploration and reduces redundant searching.
- Robots share status, exploration information, and mapping data as required by the final design.
- Communication is proposed to be implemented using ROS 2 Topics over DDS/RTPS

# Adopted Design Methodology







**1. Literature Review and Requirements Analysis**

establish context, scope, and system requirements.

**2. Simulation Environment and System Design**

define the simulated environment and proposed system structure.

**3. Implementation and Integration**

connect mapping, exploration, communication, and path planning.

**4. Testing, Performance Evaluation, and Analysis**

assess system behavior against the project objectives.

# Project Timeline & Milestones

## Project Duration: 03 Oct 2026 – 19 Jan 2027

